

NHH



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BAN402 Decision Modelling in Business

Project 2

Candidate numbers: 39 and 115

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PART A

Task 1: Which one of the 11 model formulations would you select?:

The problem of finding a minimal set of hubs such that every country can be reached within a time $\bar{t}$ seems to be a classic "coverage" problem, and we find the ***Set Covering*** model to be fitting here, as that formulation is focused on a minimum coverage while minimizing costs.

How would you use the given data to set the values of the parameters?: The cost c_i would be used as a parameter of costs to establish a hub in each of the 18 countries. These costs would be implemented in the objective function where $\sum_{i \in I} c_i \cdot x_i$, where x_i is a binary variable 1 if hub i is selected, 0 otherwise. The flight time parameter $t_{i,j}$ would be used in a constraint, where all countries are reachable from a given hub within a flight time $\bar{t}$ such that $t_{i,j} \leq \bar{t}$ exists for countries j covered by hubs i .

Could this model be infeasible?: Yes, the model could become infeasible if, for instance, no single hub (or combination of hub) can cover all countries within the flight time $\bar{t}$. In such a case, the model will not find a solution that satisfies all the constraints, rendering it infeasible.

Task 2: Which one of the 11 model formulations would you select?:

In this problem we face several added costs associated with managing the network, and for a more complex model including multiple types of costs and constraints, we find that the ***Facility Location as a Pure Integer Linear Programming*** model is appropriate, as it can effectively allocate hubs and managers, while integrating both the hub establishment costs c_i and administrative costs $r_{i,j}$, and it can also handle the constraint on the maximum number of local managers per hub m_i .

How would you use the given data to set the values of the parameters?: The hub establishment cost c_i and administrative costs $r_{i,j}$ would directly mapped to the objective function for cost minimization, where $\sum_{i \in I} c_i \cdot x_i$ as before, and $\sum_{i \in I, j \in J} r_{i,j} \cdot y_{i,j}$ where $y_{i,j}$ is 1 if local manager in country j reports to hub i . The parameter maximum local managers m_i would act as an upper-bound constraint, where $\sum_{j \in J} y_{i,j} \leq m_i \cdot x_i, \forall i \in I$.

Could this model be infeasible?: Yes, the model could become infeasible for a variety of reasons: If the managerial limitation m_i is too restrictive, it might be impossible to assign all local managers to hub managers. If the available hubs cannot serve all countries within the flight time $\bar{t}$, then the model will not find a feasible solution.

PART B**Task 1: Which modication(s) introduced to help Nils?**

Nils wants to ensure that he takes at most one hard course per term. CHard defined as a subset of C (this is the case for Task 1, 2, and 3). Courses like R, FE, and NFE have already been divided by subsets, such as R, FE, we can do the same for CHard. Making sure that the selection counter for that subset of CHard is less than or equal to one for each time period t . In this task we hold everything else equal, but add some modifications.

Set $CHard \subseteq C$: Subset of hard courses in Set C

Constraint $\sum_{i:c_i \in CHard, t_i=t} x_i \leq 1 \quad \forall t \in T$

Task 2: Which modication(s) introduced to help Lisa?

Lisa is fine with taking two or more hard courses in the same term, but she doesn't want this to happen in more than two terms in total. We add a binary variable y_t , being 1 whenever Lisa takes 2 or more hard courses, from the first constraint. The second constraint sums the number of times Lisa has taken 2 or more hard courses over the entire time period, making sure this does not happen for a total of more than 2 times. Where M is a sufficiently large positive constant, in other words larger than the maximum number of courses Lisa can take. Everything else equal, and we keep the CHard subset from Task 1. For the below constraint, whenever x_i is 2 or more, v_t must be 1 rather than 0.

Variable

$$v_t \in \{0, 1\} \quad \forall t \in T$$

Constraints $\sum_{i:c_i \in CHard, t_i=t} x_i - 1 \leq M v_t \quad \forall t \in T$
 $\sum_{t \in T} v_t \leq 2$

Task 3: Which modication(s) introduced to help Abel?

Abel wants a reward of 10 units in the objective function for each term in which exactly two hard courses appear in his study plan. The point here is to create a decision variable w_t which captures whenever Abel takes exactly two courses in a term, count the number of instances, and time this by the reward of 10 in the objective function. Where w_t is 1 if Abel takes exactly two hard courses in term t and 0 otherwise. Everything else held equal, but we keep the subset CHard from Task 1.

Variable

$$w_t \in \{0, 1\} \quad \forall t \in T$$

Constraints $\sum_{i:c_i \in CHard, t_i=t} x_i \leq 2 + M(1 - w_t) \quad \forall t \in T$
 $\sum_{i:c_i \in CHard, t_i=t} x_i \geq 2 - M(1 - w_t) \quad \forall t \in T$

Objective function

$$\sum_i [(PR(c_i) + PR(m_i) - 0.00001 \cdot ss_i)x_i - 0.01y_i - 0.01z_i] + 0.001 \sum_{i:c_i \in L} t_i x_i + 10 \sum_t w_t$$

PART C

Task 1: Formulate a mixed integer linear programming model

Sets	J : Set of crude oils	SP : Set of saleable products
	B : Set of components	P : Set of blended products
	I : Set of CDUs	D : Set of depots
	K : Set of markets	T : Set of ordered periods

Parameters: *Found in the project description*

Variables

$x_{j,i,t} \geq 0, \forall j \in J, i \in I, t \in T$	: Crude oil j refined in CDU i in period t
$z_{b,t} \geq 0, \forall b \in B, t \in T$	: Component b blended in period t
$m_{i,t} \in \{0, 1\}, \forall i \in I, t \in T$	: 1 if CDU i maintenance at t , or 0
$y_{i,t} \in \{0, 1\}, \forall i \in I, t \in T$	: 1 if CDU i active at t , or 0
$IC_{j,t} \geq 0, \forall j \in J, t \in T$	: Inventory crude oil end of day t
$IB_{b,t} \geq 0, \forall b \in B, t \in T$	: Inventory component end of day t
$ID_{d,sp,t} \geq 0, \forall d \in D, sp \in SP, t \in T$	: Inventory saleable products end of day t
$u_{b,t} \geq 0, \forall b \in B, b \neq lowqfuel, t \in T$	: Transportation b to blending end of t
$o_{d,sp,t} \geq 0, \forall d \in D, sp \in SP, t \in T$	: Transportation sp to depot d at end of day t
$v_{sp,d,k,t} \geq 0, \forall sp \in SP, d \in D, k \in K, t \in T$	: Saleable sp sent to market k end of day t
$qf_{b,d,t} \geq 0, \forall b = lowqfuel \in B, d \in D, t \in T$	: 'lowqfuel' sent to depot d at end of day t

Objective: maximize Profit

$+\sum_{sp,d,k,t>0} S_{sp} \cdot v_{sp,d,k,t-1}$	Sell yesterdays units
$-\sum_{j,i,t} C_{ref,j,i} \cdot x_{j,i,t}$	Refining cost
$-\sum_{i,t} C_{Main} \cdot m_{i,t}$	Maintenance
$-\sum_{d,p,t} C_{prod,p} \cdot o_{d,p,t}$	Production cost
$-\sum_{b,t} C_{tra1} \cdot u_{b,t}$	Transport to blending
$-\sum_{b,p,t,d} qf_{b,d,t} \cdot C_{tra2,d}$	Transport lowqfuel
$-\sum_{sp,d,t} C_{tra3,d} \cdot o_{d,sp,t}$	Transport product to depot
$-\sum_{sp,d,k,t} C_{tra4,d,k} \cdot v_{sp,d,k,t}$	Transport to market
$-\sum_{j,t} C_{invi} \cdot IC_{j,t}$	Storage at refinery
$-\sum_{b,t} C_{invb} \cdot IB_{b,t}$	Storage at blending
$-\sum_{d,sp,t} C_{invp,d} \cdot ID_{d,sp,t}$	Storage at depot

Constraints

$x_{j,i} : x_{j,i,0} = 0$	$\forall j \in J, i \in I$
$IB_{b,t} : IB_{b,t} = 0$	$\forall b \in B, 1 \leq t \leq 1$
$IC_j : IC_{j,0} = 0$	$\forall j \in J$
$U_b : u_{b,0} = 0$	$\forall b \in B$
$o_{d,p} : o_{d,p,0} = 0$	$\forall d \in D, p \in P$
$Z_{b,t} : z_{b,t} = 0$	$\forall b \in B, 1 \leq t \leq 1$
$qf_{b,d} : qf_{b,d,0} = 0$	$\forall b \in B, d \in D$
$V_{sp,d,k} : v_{sp,d,k,0} = 0$	$\forall sp \in SP, d \in D, k \in K$
$CapacityConstr1_{i,t} : \sum_{j \in J} x_{j,i,t} \geq \min R_i \times y_{i,t}$	$\forall i \in I, t > 0$
$CapacityConstr2_{i,t} : \sum_{j \in J} x_{j,i,t} \leq \max R_i \times y_{i,t}$	$\forall i \in I, t > 0$
$maintenance_not_active_{i,t} : m_{i,t} + y_{i,t} \leq 1$	$\forall i \in I, t \in T$
$LastSixDaysMaintenance_i : \sum_{t=7}^{12} m_{i,t} \geq 1$	$\forall i \in I$
$force_active_cdu_t : \sum_{i \in I} y_{i,t} \geq 1$	$\forall t \in T$
$BalanceCrude_{j,t} : IC_{j,t} = IC_{j,t-1} + E_j - \sum_{i \in I} x_{j,i,t}$	$\forall j \in J, t > 0$
$TransBlending_{b,t} : u_{b,t} = \sum_{j \in J, i \in I} A_{j,b,i} x_{j,i,t}$	$\forall b \in B \setminus \{ "lowqfuel" \}, t \geq 1$
$lowqfuel_{b,t} : \sum_{d \in D} qf_{b,d,t} = \sum_{i \in I, j \in J} A_{j,b,i} x_{j,i,t}$	$\forall b = "lowqfuel", t \geq 1$
$BalanceComponents_{b,t} : IB_{b,t} = IB_{b,t-1} - \sum_{p \in P} z_{b,t} R_{b,p} + u_{b,t-1}$	$\forall b \in B \setminus \{ "lowqfuel" \}, t > 1$
$TransDepot_{p,t} : \sum_{d \in D} o_{d,p,t} = \sum_{b \in B} R_{b,p} z_{b,t}$	$\forall p \in P, b \in B \setminus \{ "lowqfuel" \}, t > 0$
$BalanceSP_{d,sp,t} : ID_{d,sp,t} = ID_{d,sp,t-1} + o_{d,sp,t-1} - \sum_{k \in K} v_{sp,d,k,t}$	$\forall d \in D, sp \in SP, t > 0$
$transmarked_{sp,d,k,t} : v_{sp,d,k,t} \leq ID_{d,sp,t}$	$\forall sp \in SP, d \in D, k \in K, t > 0$
$DemandConstraint_{sp,k,t} : \sum_{d \in D} v_{sp,d,k,t-1} \leq \delta_{sp,k,t}$	$\forall sp \in SP, k \in K, t > 0$
$InitialDepo_{sp,d} : ID_{d,sp,0} = I_{zero,sp,d}$	$\forall sp \in SP, d \in D$
$FinalDepo_{sp,d} : ID_{d,sp,12} \geq I_{final,sp,d}$	$\forall sp \in SP, d \in D$
$FinalComp_b : IB_{b,12} \geq 100$	$\forall b \in B \setminus \{ "lowqfuel" \}$
$translowqfuel_{d,b,t} : ID_{d,b,t} = qf_{b,d,t-1} + ID_{d,b,t-1} - \sum_{k \in K, sp \in SP: sp="lowqfuel"} v_{sp,d,k,t}$	$\forall d \in D, b = "lowqfuel", t > 0$

How much is the optimal profit, when do CDUs operate, and when are the maintenance routines?: Running the model we find that **Profit** is 2688960. The below table shows 1/0 when CDU[i] is/is'nt running and being maintained for time 0-12.

Periods	0	1	2	3	4	5	6	7	8	9	10	11	12
CDU1 operates $y_{i=CDU1,t}$	1	1	1	1	1	1	1	1	1	1	1	1	0
CDU1 maintain $m_{i=CDU1,t}$	0	0	0	0	0	0	0	0	0	0	0	0	1
CDU2 operates $y_{i=CDU2,t}$	1	1	1	1	1	1	1	1	1	1	1	0	1
CDU2 maintain $m_{i=CDU2,t}$	0	0	0	0	0	0	0	0	0	0	0	1	0

Task 1b: Is there unsatisfied demand for saleable product(s)? If so, why?

We know that at the beginning of the period we can only satisfy demand using the initial inventory, which is no immediate issue, as the demands from δ is 0 for times $t=0$ and 1. The earliest we can sell new lowqfuel is day $t=3$, while selling newly refined and blended saleable product may happen from day $t=4$. We produce between 70 and 700 for each CDU, and may hold back production for the last periods. We find no unsatisfied demand for distilF and super, but some for lowqfuel, premium, and regular. This matrix can be found when running the model.

Task 1c: Is there inventory of crude oils left at the end? If so, explain why.

The first table below shows the production of crude oils j at refineries i over time t . We are aware that the inflow of crude oil is quite large at $E_j = 575$ for both crude oils, and we find that the model chooses to refine most of the CrA and CrB during the period. From the operation times, we know that CDU1/CDU2 isn't running at time $t=12/11$. Refined crude oil at time t , can earliest realize income at $t+2$, so no reason to refining during the last periods, when blending inventory IB is satisfied. The second table shows the inventory IC of CrA and CrB at the refinery. We also notice that the minimum of 70 is produced in CrB at time 12, which we have to do due to one CDU having to run, even though this does not profit.

t	0	1	2	3	4	5	6	7	8	9	10	11	12
CDU1 (CrA)	0	125	125	125	133.359	506.991	548.658	501.2	238.244	231.052	575	333.333	0
CDU2 (CrA)	0	450	450	450	441.641	68.0095	26.3417	73.8	336.756	343.948	0	0	0
CDU1 (CrB)	0	575	575	575	566.641	193.009	151.342	198.8	461.756	468.948	125	0	0
CDU2 (CrB)	0	0	0	0	8.35878	381.991	423.658	260.788	19.4943	315.214	450	0	70

t	0	1	2	3	4	5	6	7	8	9	10	11	12
CrA	0	0	0	0	0	0	0	0	0	0	0	241.667	816.667
CrB	0	0	0	0	0	0	0	115.412	209.162	0	0	575	1080

Task 1d: Is there above 100 inventory left at the end? If so, briefly explain why.

We find that there is no more than 100 inventory left at IB, as having more inventory than this contributes negatively to the objective function.

t	0	1	2	3	4	5	6	7	8	9	10	11	12
lowqfuel	0	0	0	0	0	0	0	0	0	0	0	0	0
distilA	0	0	0	0	0	0	0	0	0	0	0	16.6667	100
distilB	0	0	0	0	0	0	0	0	0	0	0	66.6667	100
naphtha1	0	0	0	0	0	0	0	0	0	0	0	0	100
naphtha2	0	0	0	0	0	0	0	0	0	0	0	0	100

Task 2: Modify your model formulation for this new situation (keeping linearity).

Remember from Task 1 we defined variable $y_{i,t}$ to be binary whenever CDU i operates, we can use this in logical expressions to add the necessary fixed costs. In example, \$60,000 is added to the objective function if $y_{CDU1,t} = 1$ and $y_{CDU2,t} = 0$, the other direction with \$120,000, and \$130,000 when the product of $y_{CDU1,t} \cdot y_{CDU2,t}$ is 1.

Variable: $y_{i,t} \in \{0, 1\}, \forall i \in I, t \in T$ 1 if CDU i active at t , or 0

Objective: Maximize profit ...

$$- \sum_t [60000 \cdot y_{CDU1,t} \cdot [1 - y_{CDU2,t}] + 120000 \cdot y_{CDU2,t} \cdot [1 - y_{CDU1,t}] + 135000 \cdot y_{CDU1,t} \cdot y_{CDU2,t}]$$

How much is the optimal profit?: Running the model we find that Profit is 1354980.

When do the CDUs operate, and when do maintenance routines take place?

Periods	0	1	2	3	4	5	6	7	8	9	10	11	12
CDU1 operates	1	1	1	1	1	1	1	1	1	1	1	1	0
CDU1 maintain	0	0	0	0	0	0	0	0	0	0	0	0	1
CDU2 operates	0	0	1	1	0	1	0	0	1	1	0	0	1
CDU2 maintain	0	0	0	0	0	0	0	1	0	0	0	1	0

PART D

Task 1: Formulate a mixed integer linear programming model

Sets V : Set of cities $D \subseteq V$: Subset of operation centers K : Set of trucks

Parameters

$\text{travel_time}_{i,j} \geq 0, \forall i, j \in V$: Distance between city i and city j
 $\text{loading_time}_d \geq 0, \forall d \in D$: Loading time at operation center d
 M : Large constant, defined as $\text{card}(V) \times \text{card}(V)$

Variables

$x_{i,j,k} \in \{0, 1\}, \forall i, j \in V, i \neq j, k \in K$: 1 if truck k goes from i to j , or 0
 $y_{i,k} \in \{0, 1\}, \forall i \in V, k \in K$: 1 if city i is visited by truck k , or 0
 $f_{i,j,k} \geq 0, \forall i, j \in V, i \neq j, k \in K$: Variable to eliminate subtours
 $z_{d,k} \in \{0, 1\}, \forall d \in V, k \in K$: 1 if city d is a depot and truck k is connected to it, or 0
 $u_{i,k} \geq 0, \forall i \in V, k \in K$: Auxiliary variable
 max_tour : Length of the maximum tour

Objective: minimize max_tour

Constraints

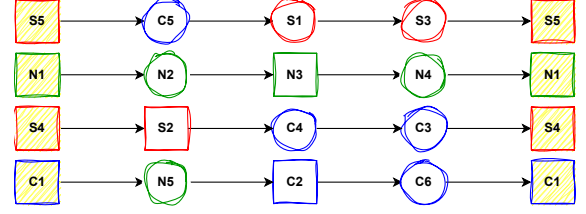
df_max_tour_k : $\text{max_tour} \geq$
 $\sum_{i,j \in V, i \neq j} \text{travel_time}_{i,j} \times x_{i,j,k}$
 $+ \sum_{d \in D} z_{d,k} \times \text{loading_time}_d \quad \forall k \in K$
 id1_k : $\sum_{i \in V} z_{i,k} = 1 \quad \forall k \in K$
 id2_k : $\sum_{d \in D} z_{d,k} = 1 \quad \forall k \in K$
 id3_d : $\sum_{k \in K} z_{d,k} \leq 1 \quad \forall d \in D$
 id4c2_upper : $\sum_{k \in K, j \in V, d \in V, j \neq d} (x_{d,j,k} - M \times (1 - z_{d,k})) \leq \text{card}(K)$
 id4c2_lower : $\sum_{k \in K, j \in V, d \in V, j \neq d} (x_{d,j,k} + M \times (1 - z_{d,k})) \geq \text{card}(K)$
 id4c3_upper : $\sum_{k \in K, j \in V, d \in V, j \neq d} (x_{j,d,k} - M \times (1 - z_{d,k})) \leq \text{card}(K)$
 id4c3_lower : $\sum_{k \in K, j \in V, d \in V, j \neq d} (x_{j,d,k} + M \times (1 - z_{d,k})) \geq \text{card}(K)$
 id5_i : $\sum_{k \in K} y_{i,k} = 1 \quad \forall i \in V$
 $\text{id6}_{k,i}$: $\sum_{h \in V, h \neq i} x_{h,i,k} = y_{i,k} \quad \forall k \in K, i \in V$
 $\text{id7}_{k,i}$: $\sum_{j \in V, j \neq i} x_{i,j,k} = y_{i,k} \quad \forall k \in K, i \in V$
 $\text{millertucklin}_{i,j,k}$: $u_{i,k} - u_{j,k} + 4 \times x_{i,j,k} \times (1 - z_{i,k}) \leq 3 \quad \forall i, j \in V, i \neq j, k \in K$

Which cities are selected as operation centres, and from which operation centre does (do) this (these) truck(s) depart?: Cities $d = \{S5, N1, S4, C1\}$ are selected as operation centers, and trucks $k = \{1, 2, 3, 4\}$ depart from those operation centers in their respective order. This is found in the model output from the binary variable $z_{d,k}$.

How much is the longest time needed by a truck in your solution?: The longest time needed by a truck in our solution max_tour is 50.88.

In your written report, outline the optimal routes that you found

Truck k	Route
1	S5-C5-S1-S3-S5
2	N1-N2-N3-N4-N1
3	S4-S2-C4-C3-S4
4	C1-N5-C2-C6-C1



Task 2: Modify your model for this new situation

Sets

V : Set of all nodes D : Subset of operation centers $\subseteq V$

K : Set of trucks H : Subset of high impact cities $\subseteq V$

C : Subset of medium impact cities $\subseteq V$

Parameters

M : Large constant

high_initial_score : Score for high impact city as initial point (default 200)

high_second_score : Score for high impact city as second city (default 160)

high_third_score : Score for high impact city as third city (default 120)

medium_initial_score : Score for medium impact city as initial point (default 150)

medium_second_score : Score for medium impact city as second city (default 100)

medium_third_score : Score for medium impact city as third city (default 50)

Variables

$x_{i,j,k}$: Binary. 1 if truck k goes from i to j , 0 otherwise

$y_{i,k}$: Binary. 1 if city i is visited by truck k , 0 otherwise

$z_{d,k}$: Binary. 1 if city d is a depot and truck k is connected to it, 0 otherwise

$u_{i,k}$: Integer. Used to eliminate sub-tours

$r_{i,k}$: Binary. 1 if city i is connected to city and that city is connected to a depot, or 0

Objective: maximize emergency_score

$$\begin{aligned}
& \sum_{i \in H, j \in V, k \in K: i \neq j} \text{high_initial_score} \times x_{i,j,k} \times z_{i,k} \\
& + \sum_{i \in C, j \in V, k \in K: i \neq j} \text{medium_initial_score} \times x_{i,j,k} \times z_{i,k} \\
& + \sum_{i \in H, j \in V, k \in K: i \neq j} \text{high_second_score} \times x_{j,i,k} \times z_{j,k} \\
& + \sum_{i \in C, j \in V, k \in K: i \neq j} \text{medium_second_score} \times x_{j,i,k} \times z_{j,k} \\
& + \sum_{j \in V, i \in H, k \in K: j \neq i} x_{j,i,k} \times \text{high_third_score} \times (1 - r_{i,k}) \times (1 - z_{i,k}) \\
& + \sum_{j \in V, i \in C, k \in K: j \neq i} x_{j,i,k} \times \text{medium_third_score} \times (1 - r_{i,k}) \times (1 - z_{i,k})
\end{aligned}$$

Constraints

$$\begin{aligned}
\text{id1}_k &: \sum_{i \in V} z_{i,k} = 1 \quad \forall k \in K \\
\text{id2}_k &: \sum_{d \in D} z_{d,k} = 1 \quad \forall k \in K \\
\text{id3}_d &: \sum_{k \in K} z_{d,k} \leq 1 \quad \forall d \in D \\
\text{id4c2_upper} &: \sum_{k \in K, j \in V, d \in V: j \neq d} (x_{d,j,k} - M \times (1 - z_{d,k})) \leq \text{card}(K) \\
\text{id4c2_lower} &: \sum_{k \in K, j \in V, d \in V: j \neq d} (x_{d,j,k} + M \times (1 - z_{d,k})) \geq \text{card}(K) \\
\text{id4c3_upper} &: \sum_{k \in K, j \in V, d \in V: j \neq d} (x_{j,d,k} - M \times (1 - z_{d,k})) \leq \text{card}(K) \\
\text{id4c3_lower} &: \sum_{k \in K, j \in V, d \in V: j \neq d} (x_{j,d,k} + M \times (1 - z_{d,k})) \geq \text{card}(K) \\
\text{id5}_i &: \sum_{k \in K} y_{i,k} = 1 \quad \forall i \in V \\
\text{id6}_{k,i} &: \sum_{h \in V: h \neq i} x_{h,i,k} = y_{i,k} \quad \forall k \in K, i \in V \\
\text{id7}_{k,i} &: \sum_{j \in V: i \neq j} x_{i,j,k} = y_{i,k} \quad \forall k \in K, i \in V \\
\text{millertucklin}_{i,j,k} &: (u_{i,k} - u_{j,k} + 4 \times x_{i,j,k} \times (1 - z_{i,k})) \leq 3 \quad \forall i, j \in V, i \neq j, k \in K \\
\text{rdef}_{i,k} &: r_{i,k} = \sum_{j \in V: i \neq j} (x_{j,i,k} \times z_{j,k} + x_{i,j,k} \times z_{j,k}) \quad \forall i \in V, k \in K
\end{aligned}$$

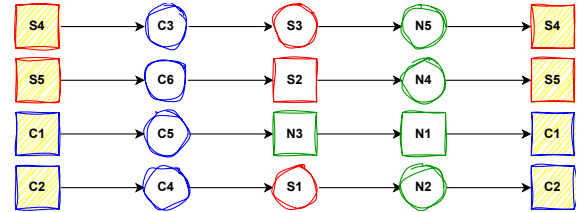
Implement and solve the model in AMPL, using the solver gurobi: We know that there are upper limits for the total amount of emergency scores one can achieve. The sum total emergency scores starting with S, and thereafter going C is: 1410 ($4 \times 200 + 160 + 3 \times 100 + 3 \times 50$), total for the first round is 800, second 640, and third 360. But this is given that all cities are possible operational cities. Also, we find that the marginal loss of S/C is 40/50 per round from 1..3, which indicates that first covering S, thereby C, and thereafter S again can be a good strategy.

What is the overall emergency score in your solution?: The `emergency_score` found in our solution is 1460.

Which cities are selected as operation centres?: Cities $d = \{S4, S5, C1, C2\}$ are selected as operation centers, and trucks $k = \{1, 2, 3, 4\}$ depart from those operation centers in their respective order. This increase of selected high and medium-impact compared to Task 1 makes sense, as this will contribute to the objective function. Looking at the figures below, there is a clear shift towards visiting the cities with prefix S and C during the first three "rounds". The somewhat interesting finding of visiting cities S during the third "round", may results from the marginal loss of S being $200 - 160 = 160 - 120 = 40$ and the marginal loss of C being $150 - 100 = 100 - 50 = 50$.

Outline the optimal trips that you found:

Truck k	Route
1	S4-C3-S3-N5-S4
2	S5-C6-S2-N4-S5
3	C1-C5-N3-N1-C1
4	C2-C4-S1-N2-C2



Task 3: What's the overall emergency score of the solution found in task 1?

Taking our solution found in task 1, we can deduce the overall emergency scores:

Truck k	Calculation	Sum
1	S5: 200 + C5: 100 + S1: 120	420
2	N1: 0 + N2: 0 + N3: 0	0
3	S4: 200 + S2: 160 + C4: 50	410
4	C1: 150 + N5: 0 + C2: 50	200
In sum		1030

Implement and solve the new model in AMPL: In this task we somewhat aggregate previous findings, using the objective function from Task 1 for minimizing the length of the longest tour, while the objective function from Task 2 is implemented through a constraints. For this purpose, formulation remains same as before, with the following modifications:

Sets: *same as before*

Parameters: optimal_emergency_score : Default value of 1460

α : For values 75%, 80%, 85%, 90%, 95%, and 100%

Variables: *same as before*

Objective: Minimize max_tour

Constraint: optimal_emergency_score $\times \alpha \leq$ emergency_score

We find that emergency score from Task 2 is emergency_score = 1460.

In your written report, summarize your results in a table with three columns showing for each of the six instances the values of alpha, the overall emergency score, and the longest time needed by a truck.

α	minimum_score	overall_emergency_score	longest_time
100%	1460	1460	75.18
95%	1387	1390	61.18
90%	1314	1340	57.46
85%	1241	1290	54.88
80%	1168	1200	50.88
75%	1095	1130	50.88